EECE.5494. MODEL-BASED SYSTEMS ENGINEERING

OOSEM, MagicGrid

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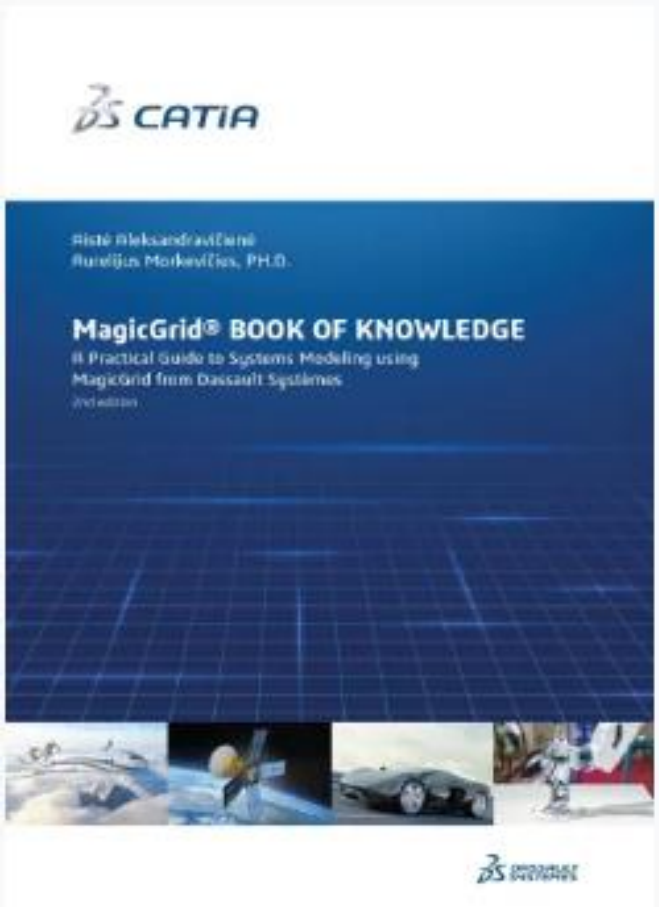
REFERENCE BOOK IN THESE SLIDES – MAGICGRID BOK



https://discover.3ds.com/magicgrid-book-of-knowledge

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WHAT IS MAGICGRID?

- MagicGrid is a methodology framework that guides the development of systems models within the CATIA No Magic suite.
- It organizes the system architecture into a series of interconnected grids, covering different aspects of the system life cycle.

	PILLAR				
DOMAIN		REQUIREMENTS	BEHAVIOR	STRUCTURE	PARAMETERS
	PROBLEM (BLACK BOX)	B1-W1 Stakeholder Needs	B2 Use Cases	B3 System Context	B4 Measurements of Effectiveness
	PROBLEM (WHITE BOX)		W2 Functional Analysis	W3 Logical Subsystem Communication	W4 Measurements of Effectiveness
	SOLUTION	S1 System Requirements	S2 System Behavior	S3 System Structure	S4 System Parameters
		SS1 Subsystem Requirements	SS2 Subsystem Behavior	SS3 Subsystem Structure	SS4 Subsystem Parameters
		C1 Component Requirements	C2 Component Behavior	C3 Component Structure	C4 Component Parameters
	IMPLEMENTATION	I1 Physical Requirements	Software, Electrical, Mechanical		

- Structured Approach
- Consistency
- Scalability
- Integration

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WHO SHOULD USE MAGICGRID?

- **Target Audience:**
  - Systems Engineers
  - Architects
  - Analysts

- **Application Domains:**
  - Aerospace & Defense
  - Automotive
  - Healthcare
  - Telecommunications
  - Any domain where complex systems are developed.

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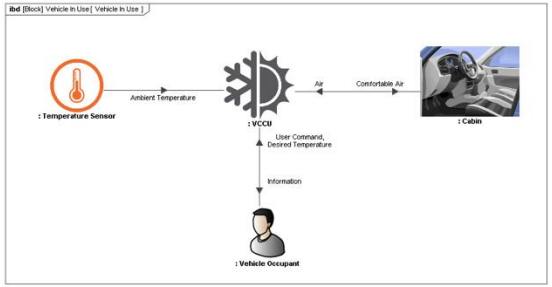
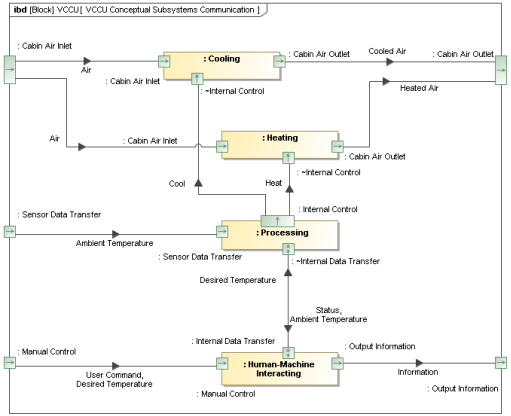
PROBLEM DOMAIN

- **Black-Box Analysis** is a method where the System of Interest (SOI) is viewed purely in terms of its inputs, outputs, and interactions, without considering its internal workings. The purpose is to understand what the SOI must do to meet stakeholder expectations.



- **White-Box Analysis** is a detailed examination of the internal workings of a system or component. Unlike black-box analysis, which focuses on the external behavior of the system (inputs, outputs, and interactions with other systems or users), white-box analysis looks inside the system to understand its internal structure, logic, and data flow.



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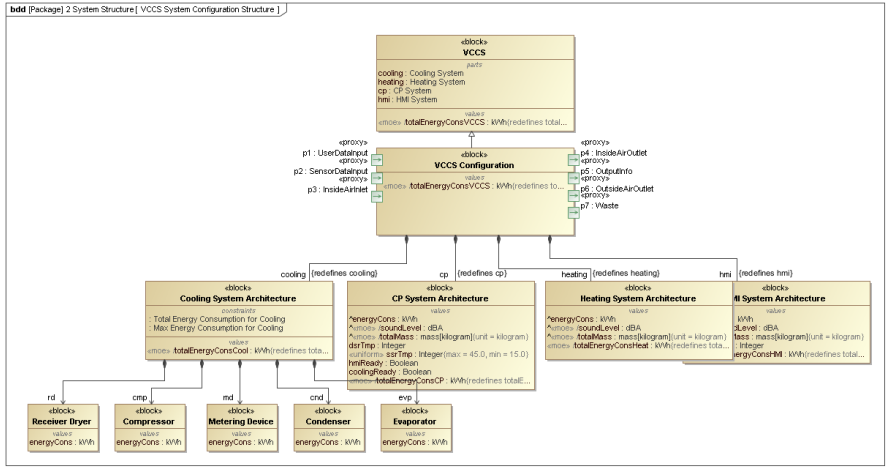
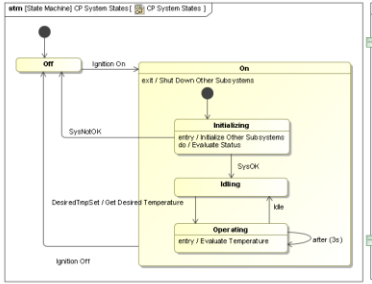
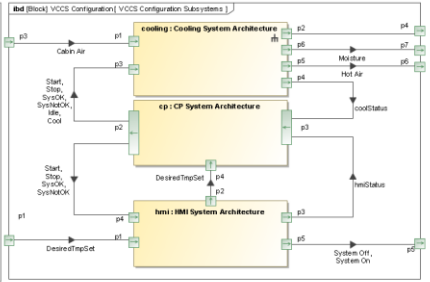
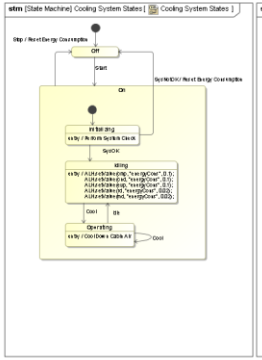
SOLUTION DOMAIN

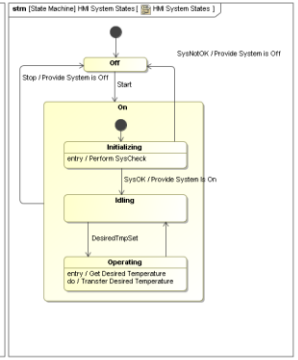
- The **Solution Domain** in MagicGrid focuses on designing and specifying the architecture that will meet the requirements defined in the Problem Domain (both Black Box and White Box). This includes detailing the logical and physical components of the system and how they work together to form a complete solution.

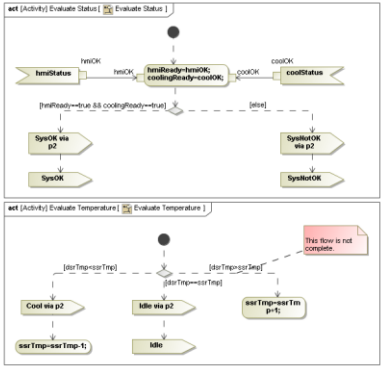












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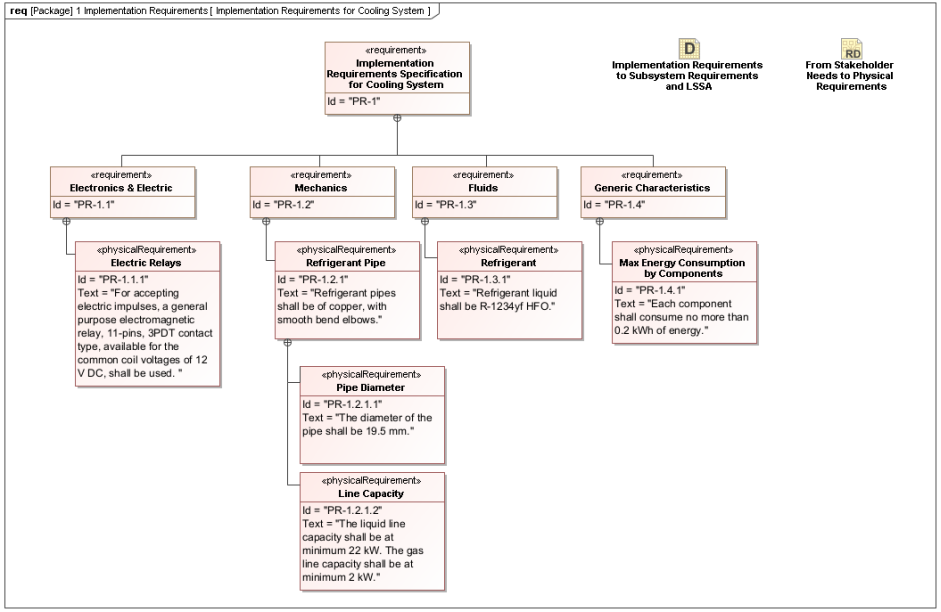
IMPLEMENTATION DOMAIN

- The Implementation Domain in MagicGrid focuses on the actual realization of the system as designed in the Solution Domain. This includes translating the architecture into working components, deploying the system, and verifying that it meets all requirements.



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REQUIREMENTS, STRUCTURE, BEHAVIOR, AND PARAMETERS

- **Requirements** are the foundation, driving the design and development process across all domains.
- **Structure** evolves from understanding internal components (Problem Domain) to designing a coherent system architecture (Solution Domain) and finally realizing it physically (Implementation Domain).
- **Behavior** is analyzed for external and internal actions (Problem Domain), formalized into models (Solution Domain), and implemented in the final system (Implementation Domain).
- **Parameters** guide design constraints and are validated throughout the development process, ensuring that the system operates within the defined limits.

This alignment ensures that every aspect of the system is thoroughly considered from problem identification through solution development and final implementation.

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